

Module 2 | Python Programming

Technical Workouts

1. Understand Python Fundamentals
 - a. Variables and data types
 - b. Input and output operations
 - c. Type conversion
 - d. Expressions and operators
 - e. Python coding conventions
2. Understand Control Flow
 - a. Conditional statements
 - b. Nested conditions
 - c. Loops
 - d. Control statements
 - e. Logical problem solving
3. Understand Collections
 - a. Lists
 - b. Tuples
 - c. Sets
 - d. Dictionaries
 - e. Collection manipulation
4. Understand Functions
 - a. Function creation
 - b. Parameters
 - c. Return values
 - d. Scope
 - e. Reusable logic
5. Understand File Handling
 - a. Reading files
 - b. Writing files
 - c. CSV files
 - d. JSON files
 - e. Processing structured data

6. Understand Exception Handling

- a. try
- b. except
- c. finally
- d. Custom exceptions
- e. Writing robust programs

7. Understand Modules & Packages

- a. Importing modules
- b. Creating reusable modules
- c. Package organization
- d. Virtual environments
- e. Dependency management

8. Develop Problem Solving Skills

- a. Number-based problems
- b. Collection-based problems
- c. String manipulation
- d. Data processing exercises
- e. Algorithmic thinking

9. Thinking with AI

- a. Use AI for:
 - i. Code explanation
 - ii. Debugging
 - iii. Refactoring suggestions
 - iv. Learning Python concepts
- b. Validate generated code
- c. Compare multiple implementations
- d. Identify inefficient solutions

10. Apply Through Hands-on Tasks

- a. Build utility programs
- b. Process files and datasets
- c. Implement reusable functions
- d. Solve Python logical problems

- e. Build mini data-processing applications
- f. Refactor existing solutions